

CBZ INSTITUTE OF EDUCATION

Physics • Class 12 • Semester - III

Full Marks: 35 | Time: 1 Hour 15 Minutes

1. In S.I system the dimension of χ (electrical susceptibility in vacuum) is,

[A] $M^{-1} L^{-3} T^4 I^2$

[B] $M^{-1} L^{-3} T^4 I$

[C] $M^{-1} L^{-3} T^4 I$

[D] $M^{-1} L^{-3} T^4$

2. The capacitance of a conducting sphere is C . The sphere is kept inside a hollow grounded sphere isocentric. The ratio of their radius is $n:(n-2)$. Now the capacitance of the sphere will be

[A] n

[B] $2n$

[C] $2/n$

[D] $n/2$

3. The potential difference between two electrodes of an electric cell of internal resistance r and e.m.f E is

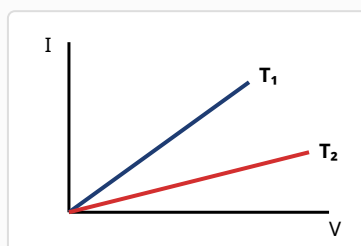
[A] $V = E + Ir$

[B] $V = E - ir$

[C] $V = Ir$

[D] $V = E \cdot Ir$

4. The I - R curve for an electrical wire for two different temperature are depicted in plot below. Find the correct alternative.



[A] $T_1 = T_2$

[B] $T_1 > T_2$

[C] $T_1 < T_2$

[D] $T_1 = 2T_2$

5. The null point of a potentiometer is shifted when current is passed for long time, because of

[A] Current flow

[B] Increase of resistance due to Joule heating.

[C] Increase of e.m.f of supply cell.

[D] Decrease of e.m.f of supply cell.

6. In Biot-Savart law the length element dl is considered instead of total length l , because

[A] for each such element the magnetic intensity changes at observation point

[B] from each such element the distance of observation point is different.

[C] condition for current flow differs at different point of conductor.

[D] all three options are correct.

7. Maglev train works on the principle of

[A] electro-magnetic field created by electromagnet.

[B] weak electromagnetic field.

[C] strong magnetic field.

[D] None.

8. Which of the alternative is correct answer after matching the columns.

Column I	Column II
Ammeter	(i) Alternating voltage step up
Voltmeter	(ii) Used in series connection
Dynamo	(iii) Closed coil is rotated in magnetic field.
Transformer	(iv) Internal resistance is very high

[A] (ii)-(iv)-(iii)-(i)

[B] (i)-(ii)-(iii)-(iv)

[C] (iv)-(iii)-(ii)-(i)

[D] (i)-(iii)-(ii)-(iv)

9. Which of the alternative is correct answer after matching the column

Column I	Column II
AM Band	(i) 54MHz
Short Wave	(ii) 54-89 MHz
Television	(iii) 530-1710 kHz
FM	(iv) 88-108MHz

[A] (ii)-(iv)-(iii)-(i)

[B] (iii)-(i)-(ii)-(iv)

[C] (iv)-(iii)-(ii)-(i)

[D] (i)-(iii)-(ii)-(iv)

10. Two equal point charges of $1\mu C$ each are located at the point $(\hat{i} + \hat{j} + \hat{k}) m$ and $(2\hat{i} + 3\hat{j} + \hat{k})$. What is the magnitude of characteristics force between them?

[A] $10^{-3} N$

[B] $10^{-12} N$

[C] $10^{-6} N$

[D] $10^{-9} N$

11. A charge Q is located at the centre of a cube. The electric flux through any surface is

[A] $2\pi Q / 6(4\pi\epsilon_0)$

[B] $4\pi Q / 6(4\pi\epsilon_0)$

[C] $\pi Q / 6(4\pi\epsilon_0)$

[D] $Q / 6(4\pi\epsilon_0)$

12. When n number of resistance each of resistance r ohm when connected in parallel gives equivalent resistance R ohm. If these resistances were connected in series the combination would have a resistance in ohms, equal to

[A] R / n^2

[B] R / n

[C] $n R$

[D] $n^2 R$

13. A conducting circular loop of radius r carries a constant current I . It is placed in an uniform magnetic field $B \rightarrow$, such that $B \rightarrow$ is perpendicular to the plane of the loop. The magnetic force acting on the loop is,

[A] $I r B \rightarrow$

[B] $\pi r I$

[C] 0

[D] $2\pi r I B \rightarrow$

14. Two charges are placed at a certain distance apart. A metallic sheet is placed between them. What will happen to the magnitude of the force between the charges?

- [A] Increase [B] decrease
[C] Remain same [D] Either [A] or [B]

15. To send 10% of the main current through a moving coil Galvanometer of resistance 99 ohm, the shunt required is,

- [A] 9 ohm [B] 10 ohm
[C] 11 ohm [D] 9.9 ohm

16. Magnetic field produced by atoms and molecules is due to type of motion of electrons,

- [A] neither spin nor orbital motion [B] spin motion only
[C] spin and orbital motion both. [D] orbital motion only.

17. A magnetic needle lying parallel to the magnetic field requires W unit of work to turn it around 60° . The torque needed to maintain the needle in this position will be,

- [A] $(\sqrt{3}/2) W$ [B] $2 W$
[C] $\sqrt{3} W$ [D] W

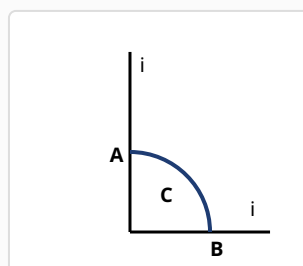
18. A radio set operates at 3 volt D.C. A transformer with 12 turns in secondary coils used to step-down the input 220 V.A.C e.m.f to 3V.A.C e.m.f. This A.C e.m.f is rectified by another circuit to give 3Volt D.C to feed into the radio. The number of turns in the primary coil is

- [A] 36 [B] 330
[C] 1 [D] 880

19. The back e.m.f in a d.c motor is maximum, when the motor has

- [A] minimum speed. [B] just started moving.
[C] just switched off. [D] picked up maximum speed.

20. A wire carrying current i is shaped as shown in figure. Section AB is a quarter circle of radius r . The magnetic field is directed



- [A] Perpendicular to the plane of the paper and out of it [B] At an angle to the plane of the paper.
[C] Along the bisector of the angle ACB away from AB [D] Along the bisector of the angle ACB towards AB.

21. If a current is passed through a metallic spring, then the spring will

- [A] Compress [B] Expand
[C] Remain same [D] None of these.

22. When a body is charged by induction, then the body

- | | |
|--------------------------------------|-------------------------------------|
| [A] Becomes neutral | [B] Does not loose any charge. |
| [C] Loses whole of the charge in it. | [D] Loses part of the charge in it. |

23. The electromagnetic wave does not transport

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|--------------|------------------|
| [A] energy | [B] charge |
| [C] momentum | [D] information. |

24. A sensitive magnetic instrument that be shielded very effectively from outside magnetic field by placing it inside a box of

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|--------------------------|--|
| [A] diamagnetic material | [B] metal box with high conductivity. |
| [C] glass box | [D] soft iron box with high permeability |

25. A 200 Volt 50 Hz main is applied to LCR ckt, across which of those components, the potential difference is much higher than 200 Volt.

- | | |
|----------------|---|
| [A] Resistance | [B] Capacitance |
| [C] Inductance | [D] Capacitance and inductance depending upon the value of C and L. |

26. An aeroplane of wing span of 25 m is flying at horizontal speed of 100 m / s in a region Where, the vertical component of the earth's magnetic field is $5 \times 10^{-5} \text{ Wb/m}^2$. Find the potential difference between the tips of the wings

- | | |
|-----------------------|-----------------------|
| [A] 0.125 V | [B] 0.152 V |
| [C] 0.15 V | [D] 1.25 V |

27. An e.m.f 1.5 kv is induced in a coil when a current of 4 amp collapses uniformly to zero in 8 ms. The inductance of the coil is

- | | |
|----------------------------------|--------------------|
| [A] $7.5 \times 10^5 \text{ Hz}$ | [B] 3Hz |
| [C] 0.3Hz | [D] 3.3Hz |

28. The temperature co-efficient of resistance of a metal wire is $0.00125/^\circ\text{C}$. Its resistance is 1ohm at 300K. Its resistance would be 2ohm at,

- | | |
|--------------------|---------------------|
| [A] 1154K | [B] 1100K |
| [C] 110K | [D] 110.1K |

29. Electric current is a

- | | |
|------------------------|---------------------|
| [A] scalar quantity | [B] vector quantity |
| [C] unitless quantity. | [D] none of these. |

30. The earth's magnetic field always has a horizontal component except at

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|-----------------------|-----------------------|
| [A] equator | [B] magnetic pole |
| [C] a latitude of 600 | [D] a latitude of 500 |

31. The lines of force due to Earth's horizontal magnetic field are

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|----------------------------|-------------------------|
| [A] parallel and straight. | [B] concentric circles. |
| [C] elliptical . | [D] curved lines. |
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32. The time period of oscillation of a freely suspended bar magnet on earth with the usual notations given by

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|--------------------------------|---------------------------------|
| [A] $T = 2\pi \sqrt{(M / H)}$ | [B] $T = 2\pi \sqrt{(1 / MH)}$ |
| [C] $T = 2\pi \sqrt{(1M / H)}$ | [D] $T = 2\pi \sqrt{(M / I H)}$ |
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33. State which one of the following is correct?

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|----------------------------|----------------------------|
| [A] joule = coulomb × volt | [B] joule = coulomb / volt |
| [C] joule = volt / ampere | [D] joule = volt × ampere |
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34. As one penetrate an uniformly charged sphere, the electrical field strength E

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|------------------------------------|----------------------------|
| [A] increase | [B] decrease |
| [C] remains same as at the surface | [D] is zero at all points. |
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35. No current flows between two charged bodies connected together when they have the same

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|-----------------|----------------|
| [A] charge | [B] potential |
| [C] capacitance | [D] resistance |
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